Asher Straus

624 Ontario Ave, Tampa, FL 33606 | asherstraus2027@u.northwestern.edu | (914) 924-6220
Portfolio: **strausasher.github.io/Portfolio** | LinkedIn: **linkedin.com/in/asher-straus-0330452a9**

EDUCATION

Northwestern University | McCormick School of Engineering | Evanston, IL — Sep 2023 – Mar 2028 (expected)
B.S. Manufacturing & Design Engineering | *Segal Design Certificate* | *Robotics Certificate* | *Minor in Art*
GPA: 3.6 / 4.0 cumulative · 3.9 / 4.0 since sophomore year

WORK EXPERIENCE

Siemens Healthineers — *Mechanical Engineering Co-op, Molecular Imaging · Hoffman Estates, IL* — Jun 2026 – Present
- Updated CAD models and drawings for mechanical components in SPECT and PET scanners to resolve interferences and improve manufacturability; applied GD&T to fully define part requirements.
- Designed fixtures and brackets in Siemens NX/Designcenter, using 3D-printed prototypes to evaluate and iterate on each design.
- Contributed to the development of a multi-axis stage and investigated motorized drives to support the SPECT research team.

P1 Motor Club — *Virtual Test Driver, Research & Promotions Intern · Tampa, FL* — Apr 2023 – Present
- Test new track layouts in simulators and give performance feedback to engineers to inform design changes.
- Built a large, mostly 3D-printed model of the track from the civil engineer's Civil 3D files as a portable membership sales tool.

MIT Beaverworks Summer Institute — *Teaching Assistant, "Basics of ASICs" · Cambridge, MA* — Jun 2024 – Aug 2024
- Guided students through ASIC schematic design, simulation, and testing, and helped refine labs and curriculum.

Monticello Motor Club — *Go-Kart Instructor, Technician & Social Media Coordinator · Monticello, NY* — Jun 2018 – Jul 2022
- Diagnosed and repaired kart systems for safe, reliable operation; coached drivers of all skill levels.

DESIGN & PROTOTYPING PROJECTS

WaterGuard: CT Scanner for Aquatic Animals — *Design & Build Engineer · Major Aquarium · Chicago, IL* — Sep 2025 – Mar 2026
- Designed a removable, reusable LDPE liner and runoff collector that waterproofs a ~$1M CT scanner used to image aquatic animals.
- Led liner geometry: an oblique frustum with a 12.5° runoff slope inside 6″ of table clearance, set by full-scale wrinkled-liner tests.
- Tensile-tested heat-welded seams (n = 5): 94.8 N dry vs. 106.4 N after 24 h in saltwater, with no weld separation; co-wrote the FMEA.
- On-site tests with veterinary staff: setup fell from 45+ min to under 4. Their feedback on laser visibility drove a clear PETG window.

ExtendIt Desk Extension — *Designer & Engineer* — 2024
- Ran 22 user interviews and tested 7 mockups with 10 users each; 8 of 10 preferred the final locking desk extension (+50% surface).
- Built a load-test rig: held 33.5 lb at 3/8″ deflection against a 30 lb / 0.5″ target. Costed at $10.19 per unit for sale to universities.

Bionic Wrench Assembly Line — *Designer & Manufacturing Engineer · DSGN 386* — Mar 2025 – Jun 2025
- Reverse-engineered a commercial wrench (calipers, micrometers, optical comparator); led track design across 7 fixture versions.
- Mapped the assembly flow and cut it to a single operator; a 30-minute pilot run hit a 3:58 cycle time.
- Wrote the SOP, FMEA and value stream map; planned scale-up to 1M units/yr: 25 parallel cells, 6 s peak takt, $10.22/unit, ~$1.2M line.

Performance Bicycle Brake Caliper — *Simulation-Driven Design* — 2025
- Turned rider needs into specs (≤70 N input, ≤11 mm deflection, 15 m stop, ISO 4210); cut mass with FEA and topology optimization.
- SLS Nylon-12 prototype passed every metric on the bike: 41.3 g, 11.3 m stop at 68.9 N input, $8.26 per part.
- Led the stiffness–mass trade-off: a 14% lighter second version needed 102 N to brake, so we kept the first.

SME Dorm Accessibility Project — *Team Lead · SME National Student Manufacturing Innovation Challenge* — Aug 2026 – Present
- Lead a team of 5 building a dorm accessibility prototype for SME's April 2027 national contest; helped organize NU's two teams.
- First concept: a door-mounted knock-to-light alert (ESP32-C3, accelerometer knock-vs-slam detection), ~$50 vs. $262 commercial.

Assistive Sensory Devices — *Project Manager · Misericordia* — 2024 – Mar 2025
- Designed three sensory devices for adults with disabilities (each under $150); resident testing drove changes like larger knobs.

Formula SAE Brake System — *Suspension Member · Northwestern Formula Racing* — Sep 2024 – Present
- Sized brakes for 1 G deceleration (~42 lbf pedal to ~690 lbf tire force) using an analytical model and an improved team MATLAB sim.
- Designed rotors and checked stress and heat in Ansys, then machined them, planned line routing, and installed and bled the system.

Advanced Intelligent Manufacturing Lab — *Researcher · NSF ERC-HAMMER · MSEC 2026 Poster Co-author* — Nov 2025 – Apr 2026
- Co-authored a poster on an AI sheet-metal bend-planning pipeline; built a SolidWorks VBA tool to auto-generate its test drawings.

TECHNICAL SKILLS

Design & Prototyping: SolidWorks, Siemens NX/Designcenter, Onshape, AutoCAD, GD&T, Engineering Drawings, Rapid Prototyping (3D Printing, Laser Cutting, Waterjet), Mockups & Functional Prototypes, Fixture & Jig Design, Snap-Fit/Compliant Design, DFM/DFR

Fabrication & Test: Machining (Lathe/CNC/Mill), Welding, Sand/Iron Casting, Sheet Metal, Metrology, Tensile Testing, ESP32/Raspberry Pi

Analysis: FMEA, Ansys FEA/Topology Opt., MATLAB, Python, VBA/SolidWorks API, Cost Modeling, Lean/VSM, Line Balancing

User-Centered Design: User Interviews, Usability Testing, Concept Iteration, Human-Centered Product Design (Segal Design Certificate)

Clubs: SME Student Chapter (Secretary), Formula SAE, Baja SAE, Combat Robotics